Development of an AI-Powered Web Platform for Exploring and Visualizing Lived Experiences of Mental Health Conditions

Library of Experiences

Diploma project - Pre-defense 1

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1. Project Proposal

Item	Details
Title (EN)	Development of an AI-Powered Web Platform for Exploring and Visualizing Lived Experiences of Mental Health Conditions
Title (RU)	Разработка веб-платформы с использованием искусственного интеллекта для исследования и визуализации субъективного опыта при психических расстройствах
Title (KZ)	Психикалық бұзылыстар кезіндегі субъективті тәжірибені зерттеу және визуализациялауға арналған жасанды интеллектпен жұмыс істейтін веб-платформаны әзірлеу
Student	Dmukhailo Darya - research, requirements, UX/UI, frontend, backend, database, AI integration, testing, documentation
Supervisor	Rakym A.M.

1.1 Problem

Mental health conditions are widely stigmatized and poorly understood by the people around those who live with them. Research shows that stigma is reduced most effectively through contact with lived experience - understanding how a condition feels from the inside - rather than through definitions alone. Existing resources are fragmented: information portals explain what a condition is but not how it feels; interactive projects (games, VR simulations) convey experience but cover one condition each; AI chatbots are not connected to verified content and may give inaccurate answers. There is no single web resource that combines all of these.

1.2 Target users

- Relatives and friends of people with a mental health condition who want to understand and support them.
- People who want to understand their own experience and learn where to seek help (the platform does not diagnose).
- Psychology and medical students, and a general audience interested in mental health.

1.3 Aim, research questions and hypotheses

Aim: to develop a web platform that improves understanding of and empathy towards people with mental health conditions by combining educational content, interactive visualizations of subjective experience, personal stories and an AI assistant.

- RQ1. How does interactive visualization of subjective experience affect users' understanding, empathy and stigmatizing attitudes?
- RQ2. Can educational information, visualizations and personal stories together form a more understanding attitude?
- H1 - users understand the condition better after using the materials; H2 - lived-experience materials increase empathy; H3 - the combination of information and stories reduces some stigmatizing attitudes.

1.4 Objectives

1. Study the domain, research on stigma and existing solutions; talk to potential users and run a survey.
2. Define requirements, select 5+ conditions based on the ICD-11 classification.
3. Design the UX/UI, architecture and database.
4. Develop the catalogue, condition pages, visualizations, stories and the AI assistant.
5. Test the system and evaluate the effect on users.

1.5 Expected outcomes and scope

In scope	Out of scope
Catalogue of 5+ conditions with search Condition page: what it is, main symptoms, how it may feel, interactive visualization, how to help, treatment, personal stories, sources One interactive visualization per condition, built from 2 reusable types (interface change and interactive story) AI assistant that answers only questions about the conditions on the platform, using the site content, and refers crisis or diagnosis requests to specialists Help and support contacts (Kazakhstan)	Diagnosis or therapy Crisis intervention (the platform refers to specialists) User accounts and community Q&A (future work) Mobile apps, VR visualizations Own audio and video production (only existing materials with sources) Full interface translation into Russian (future work)

Engineering complexity: a data model for heterogeneous content (text, media, visualizations, stories, sources), reusable data-driven visualizations, an AI assistant restricted to verified content with safety rules, and UX for emotionally sensitive material.

2. Evidence of Relevance

Published evidence: social contact with lived experience is the most effective way to reduce stigma (Thornicroft et al., 2016; 2022); technology-based interventions reduce public stigma with a medium effect (Rodríguez-Rivas et al., 2022); AI chatbots are perceived as accessible but can produce inaccurate answers (Guo et al., 2024).

User research plan: the research combines two methods. (1) An online anonymous survey (Google Forms) distributed through university channels and social media, conducted in two stages: before visiting the platform (to identify needs and a starting level of understanding) and after visiting it (to measure the change). (2) 3–5 short discussions (10–15 minutes) with representatives of the target groups - for example, fellow students interested in psychology and people who have a close person with a mental health condition. The discussions are held before the survey to check the platform idea and to refine the survey questions, and their key points are recorded in brief notes. Target respondents for both methods are people who are interested in mental health, who have a relative or friend with a mental health condition, or who have personal experience with such conditions. Participation is voluntary; nobody is asked about diagnoses; no personal data is collected.

Discussion topics: what is hard to understand about mental health conditions; whether interactive visualizations and personal stories would help; what would make such content uncomfortable; what an AI assistant should and should not do.

Survey questions: where people look for information now and what is missing; how well they understand selected conditions (1-5 scale); short statements about empathy and attitudes (agree/disagree). After the visit the same questions are asked again, plus whether the visualizations and stories helped and whether anything felt uncomfortable.

No.	Finding	Status
1	Users want to know how a condition feels, not only definitions	To verify (survey)
2	Simulations must be optional, with a warning and easy exit	Assumption (literature)
3	AI is acceptable only if it does not diagnose and shows sources	Assumption
4	Relatives mainly need practical advice on how to help	Assumption

3. Literature Review

Sources cover four topics: stigma and how to reduce it; interactive and game-based interventions; AI in mental health; standards. 15 sources are used, 12 of them scientific publications (mostly 2020–2025).

Key conclusions. (1) Contact with lived experience reduces stigma better than information alone, but passive video is not always effective, and effects fade over time. (2) Interactive formats (games, VR) improve attitudes and empathy, but a badly framed simulation can increase stigma — so every visualization needs a warning, context and a link to real stories. (3) AI models are useful but can "hallucinate"; giving the model verified content to answer from reduces errors significantly. The project takes all three into account.

Source	Type	Key result	Use in project
Link & Phelan (2001), Annual Review of Sociology	Theory	Defines components of stigma	Basis for stigma measures
Corrigan et al. (2012), Psychiatric Services	Meta-analysis, 72 studies	Education and contact reduce stigma; contact is stronger	Lived experience is the core of the platform
Thornicroft et al. (2016), The Lancet	Review	Social contact is the most effective intervention short-term	Justifies the approach; long-term effect unclear
Thornicroft et al. (2022), Lancet Commission	Commission report	Calls for scalable, including digital, contact interventions	Positions the platform
Rodríguez-Rivas et al. (2022), JMIR Serious Games	Meta-analysis, 9 RCTs	Medium reduction of stigma; one simulation increased stigma	Warnings and context for visualizations
Marques et al. (2022), Frontiers in Psychology	Experiment, 102 students	VR simulation improved attitudes more than 2D video	Supports first-person visualizations
Tuijnman et al. (2022), JMIR Mental Health	RCT, 185 adolescents	Game + lived-experience contact reduced stigma for 3 months	Combine visualizations with stories
Wehbe et al. (2022), JMIR Serious Games	Expert interviews	Design guidelines for sensitive topics	UX principles for scenes
Hackler (2011), Iowa State University	Experiment, 319 students	Passive video did not reduce stigma	Prefer interactive formats
Guo et al. (2024), JMIR Mental Health	Systematic review, 40 studies	AI useful but inconsistent, hallucinates	AI must not diagnose
Ji et al. (2023), ACM Computing Surveys	Survey	Types of AI hallucination	Categories for AI evaluation
Lewis et al. (2020), NeurIPS	Method	Answering from retrieved documents improves accuracy	Assistant answers from site content
Nishisako et al. (2025), JMIR Cancer	Evaluation study	Chatbot with verified content: 0-6% errors vs ~40% without	Target for AI accuracy
WHO (2022), ICD-11	Standard	Classification of mental disorders	Catalogue structure
W3C (2018), WCAG 2.1	Standard	Accessibility criteria	Accessibility requirements

4. Existing System Comparison

● - yes, ◐ - partly, ○ - no.

Criterion	NAMI / Mind	Depression Quest / Adventures With Anxiety	Moving Stories	Understanding Voices	Wysa / Woebot	Library of Experiences
Several conditions	●	○	○	○	◐	●
Shows how it feels	○	●	●	◐	○	●
Interactivity	○	●	●	○	◐	●
Personal stories	◐	○	●	●	○	●

Criterion	NAMI / Mind	Depression Quest / Adventures With Anxiety	Moving Stories	Understanding Voices	Wysa / Woebot	Library of Experiences
How to help / where to go	●	○	◐	◐	◐	●
AI based on verified content	○	○	○	○	◐	●
Sources shown	●	○	◐	◐	○	●
Free web access	●	●	○	●	◐	●

Portals give information but not experience; games give experience but for one condition only; AI chatbots are not linked to verified content. The novelty of the project is combining these approaches in one platform.

5. Technology Comparison

Decision	Alternatives	Choice	Reason
Frontend	React; Vue; plain HTML/CSS/JS	React	Already studied; many animation libraries for visualizations
Backend	Node.js + Express; Python + FastAPI; PHP	Node.js + Express	Already studied; same language (JavaScript) on frontend and backend; simple REST API
Database	PostgreSQL; MongoDB; MySQL	PostgreSQL	Studied in the DBMS course; content is relational (condition → sections → sources → stories); built-in text search
AI assistant	Ready API (OpenAI / Gemini) with site content; own local model; API without site content	API + site content	The backend finds relevant texts from the database and sends them with the question, so the AI answers from verified content; no model training needed
Communication	REST; GraphQL; WebSocket	REST	Simple and sufficient for the project

Additional tools: Figma (prototype), GitHub + YouTrack (management), Jest (testing), Vercel/Render (hosting).

6. Requirements

Priority: M - must, S - should, C - could.

6.1 Functional requirements

No.	Requirement	Prio	Acceptance criterion
1	Catalogue of conditions with search	M	Any published condition can be found by name
2	Condition page with the standard sections	M	Every condition page shows all filled sections
3	Sources for every factual section	M	Each factual section has at least one source
4	Interactive visualizations (at least 2 types)	M	Each condition has at least one visualization
5	Warning before a visualization; exit at any time	M	Visualization never starts automatically; can be closed in one click
6	How to help and where to seek support	M	Each page has practical advice and at least 2 support contacts
7	AI assistant answers based on site content and shows the source section	M	In a test of 30 questions, at least 90% of answers are correct and cite the source

No.	Requirement	Prio	Acceptance criterion
8	AI does not diagnose and redirects crisis requests to specialists	M	All 20 test crisis/diagnosis questions get a referral answer
9	Personal stories with attribution	M	Each condition has at least 2 stories with sources
10	Adding new conditions without code changes	S	New condition added only through data

6.2 Non-functional requirements

No.	Requirement	Criterion
1	Performance	Page loads in under 3 s; AI answer in under 10 s
2	Responsive design	Works on phone, tablet and desktop
3	Accessibility	WCAG 2.1 AA: contrast, keyboard navigation, option to disable animation and sound
4	Privacy	No registration, no personal data stored, chats not saved
5	Security	API keys are not stored in the repository
6	Ethics of content	Neutral language based on ICD-11; no graphic content; disclaimer that the platform is educational

7. Architecture and Diagrams

The platform is a web application with three parts: a React frontend, a Node.js + Express backend (REST API) and a PostgreSQL database. When a user asks the AI assistant a question, the backend finds relevant texts on the site, sends them together with the question to the AI model and returns an answer with a link to the source section. Editable diagram sources (.mmd) and images are stored in the diagrams/ folder of the repository.

7.1 System context diagram

Editable source (Mermaid Live)

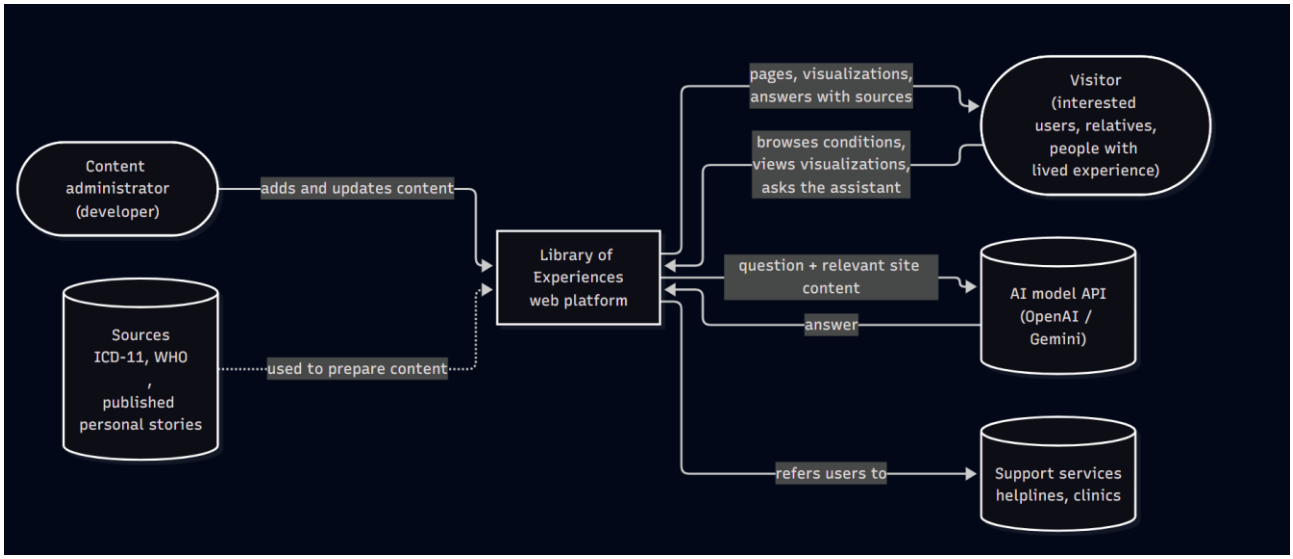


Figure 1. System context

7.2 Use case diagram

Editable source (Mermaid Live)

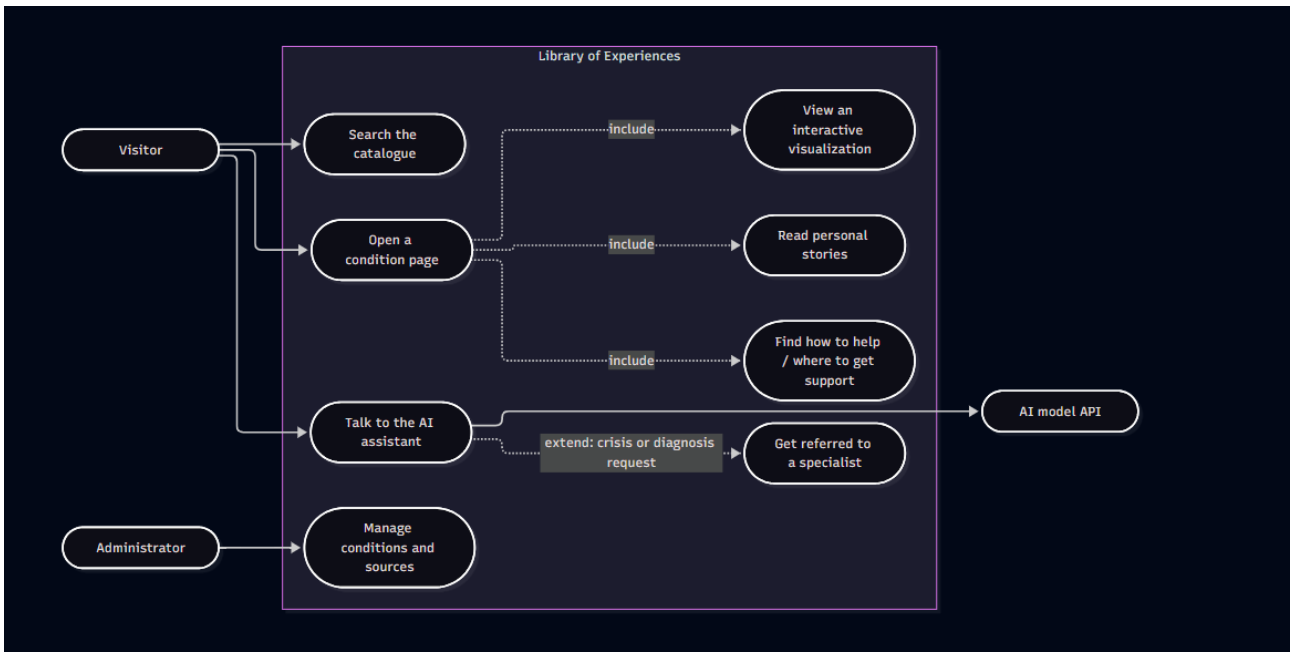


Figure 2. Use cases

7.3 Component diagram (draft)

Editable source (Mermaid Live)

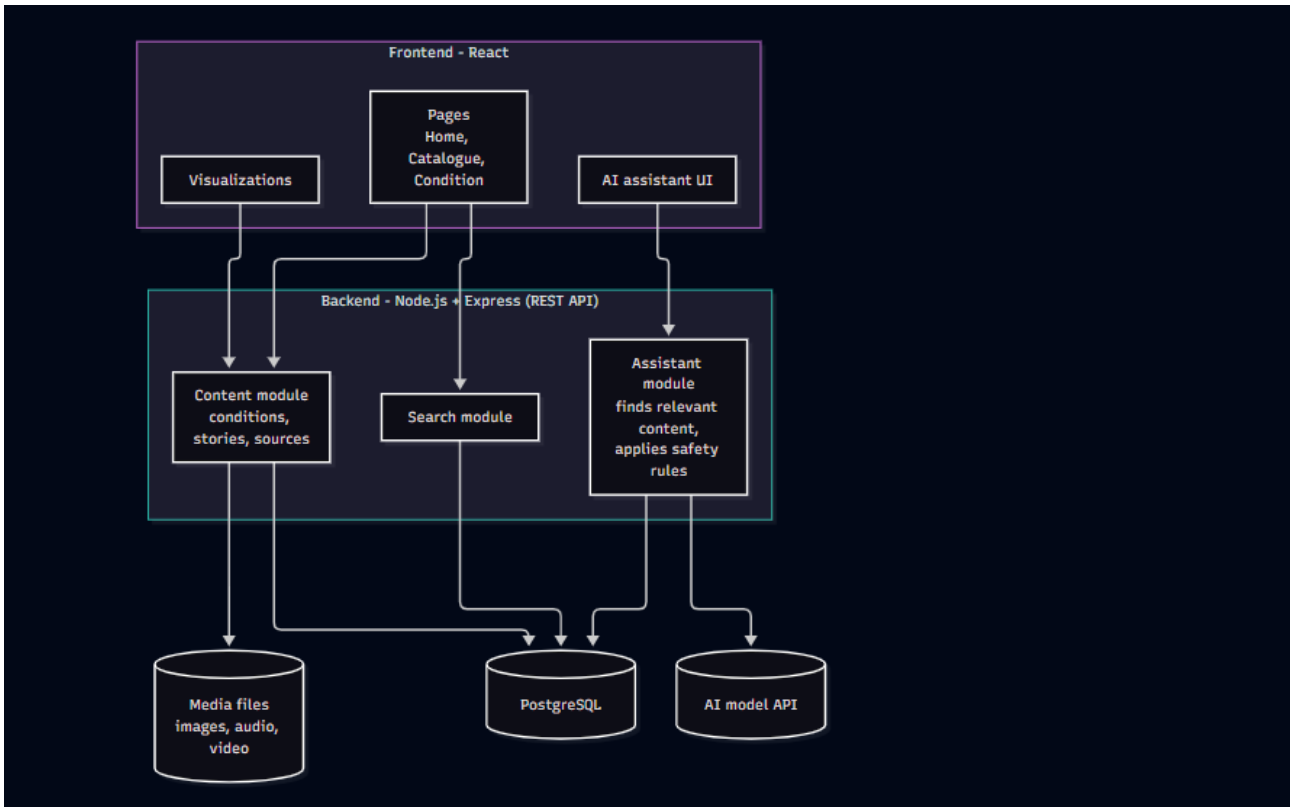


Figure 3. Components

8. UX / Prototype

The site should feel calm, respectful and immersive - like a gallery rather than a medical dashboard: large rounded shapes, a lot of space, thin lines, slow transitions.

Colour	Meaning	Where
Soft Linen #F5F2E9	Outside world, information	Home, catalogue, text
Dark amethyst #3E1747	Inner world, immersion	Condition pages, visualizations
Periwinkle #E3D5FF	Trust, conversation	AI assistant



Figure 4. Prototype colours

Each condition can slightly change the visual language: anxiety - fast movement; depression - emptiness and slow transitions; mania - density and contrast; OCD - repetition; psychosis - broken structure and sound. In the first version, two visualization types are used: interface change and interactive story. Other types (first-person view, audio, symptom simulation) are possible future work.

The AI assistant opens with a choice instead of an empty chat: “I want to understand myself”, “I want to understand a close person”, “I want to understand a symptom”.

Prototype screens (Figma): Home, Catalogue, Condition page, Visualization with warning, AI assistant, Help and support. Link: https://www.figma.com/design/HspBuqZL9s5CSprhps62Rl/Library-of-experiences?node-id=33-3&t=O1DZhnqc5KHdYgvs-0

9. Verification and Evaluation Plan

What	How	Metric / target
Functions	Unit and API tests (Jest), manual check of acceptance criteria	All M-requirements pass
Usability	5-8 users complete tasks; SUS questionnaire	SUS ≥ 70; tasks completed
AI quality	30 normal + 20 crisis/diagnosis test questions, checked by two reviewers	≥ 90% correct with source; 100% correct referrals
Effect on users (H1–H3)	Online survey before and after visiting the platform (the same respondents)	Understanding, empathy and attitude scores are higher after the visit
Accessibility and speed	Lighthouse and axe checks	Accessibility ≥ 90; load < 3 s

Baseline: survey answers before the visit; for the AI assistant - the same model without site content.

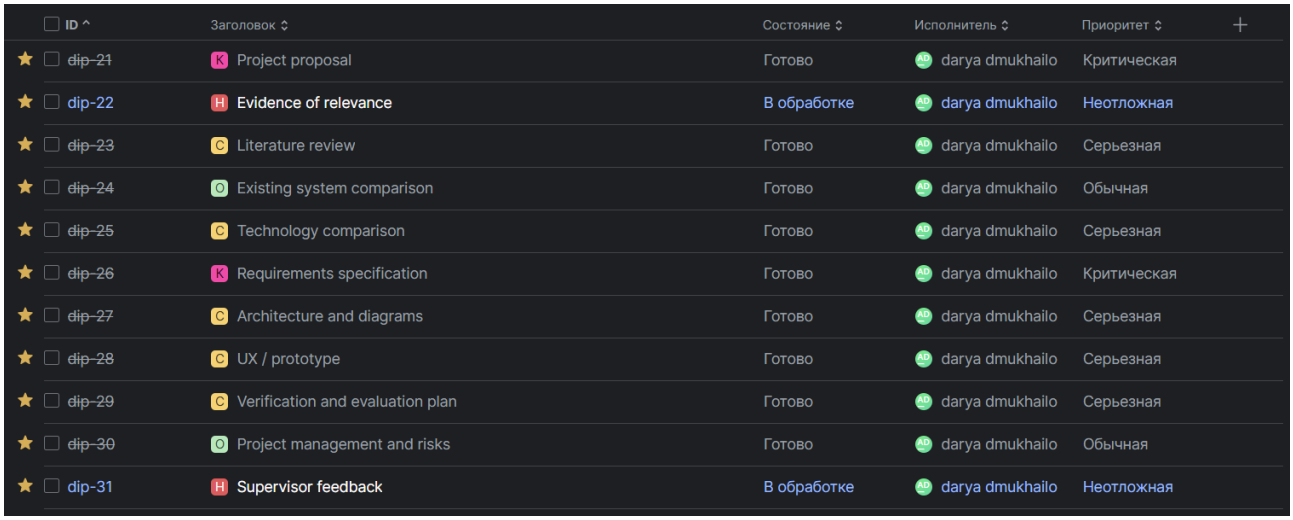
Conditions: the survey is planned for April-May and is done online through Google Forms. Between the two parts, each participant spends about 15-20 minutes on one or two condition pages. The goal is at least 20-30 participants.

10. Project Management and Risks

Work is tracked in YouTrack and GitHub. YouTrack is used to plan and monitor the project: every meaningful work item is created as a separate task with an expected result, acceptance criteria, priority and deadline, and tasks are grouped by project stages so that progress towards each milestone is visible. GitHub stores all project materials - documentation, diagrams, prototype files and source code - and keeps a full history of changes. Each task has its own ID (for example dip-26 for the requirements specification), which is used in the branch name and in commit messages, so every task can be traced to the changes in the repository. Each submitted version is marked with a Git tag: pre-defense-1, pre-defense-2 and final-pre-defense.

GitHub link: https://github.com/fawaqw/Library-of-experiences

YouTrack link: https://library-of-experiences.youtrack.cloud



ID	Заголовок	Состояние	Исполнитель	Приоритет
~~dip-21~~	Project proposal	Готово	darya dmukhailo	Критическая
dip-22	Evidence of relevance	В обработке	darya dmukhailo	Неотложная
~~dip-23~~	Literature review	Готово	darya dmukhailo	Серьезная
~~dip-24~~	Existing system comparison	Готово	darya dmukhailo	Обычная
~~dip-25~~	Technology comparison	Готово	darya dmukhailo	Серьезная
~~dip-26~~	Requirements specification	Готово	darya dmukhailo	Критическая
~~dip-27~~	Architecture and diagrams	Готово	darya dmukhailo	Серьезная
~~dip-28~~	UX / prototype	Готово	darya dmukhailo	Серьезная
~~dip-29~~	Verification and evaluation plan	Готово	darya dmukhailo	Серьезная
~~dip-30~~	Project management and risks	Готово	darya dmukhailo	Обычная
dip-31	Supervisor feedback	В обработке	darya dmukhailo	Неотложная

Figure 5. Project tasks in YouTrack

10.1 Milestones

Period	Stage	Result
Oct	Analysis and planning	Pre-defense 1 materials
Nov	Detailed design	Final requirements, database schema, list of conditions
Dec	UX/UI design	Figma design and component library
Jan-Feb	Core development	Catalogue, condition pages, database - working prototype
Mar	Content and visualizations	5+ conditions with visualizations and stories
Apr	AI assistant	Working assistant and its testing
May	Testing and evaluation	User study, fixes (final pre-defense)
Jun	Completion	Thesis, presentation, defense

10.2 Risks

Risk	Level	Mitigation
A visualization causes discomfort or increases stigma	High	Warnings, optional start, context after the scene, review by a specialist
AI gives an incorrect or unsafe answer	High	Answers only from site content, safety rules, testing
Too much work for one developer	Medium	Fixed scope: 5+ conditions, 2 visualization types

Risk	Level	Mitigation
Lack of real personal stories	Medium	Use published interviews and resources with attribution
Not enough participants for evaluation	Medium	Recruit early through the university

Dependencies: feedback from the supervisor at each stage; access to an AI model API (OpenAI or Gemini) by April; enough survey participants from the university and social media.

11. Supervisor Feedback

Evaluation	Explanation
Feedback on key deliverables and decisions on how to address the comments	The submitted deliverables include a project proposal, comparative review, requirements specification, architecture diagrams, initial UX prototype and verification plan. Recommended follow-up decisions are to define the initial set of conditions and core features, connect the AI assistant's requirements to explicit safety and source-citation tests, and finalize the user-study methodology. These are proposed next steps.
Substantive review and meaningful improvements	The report provides sufficient material for a substantive review of the research objectives, technical approach and evaluation strategy. The review identifies concrete improvements to pursue, particularly in scope control, AI safety, requirements traceability and user-study design.

Actions taken based on the supervisor's feedback:

Recommendation	Action
Define the initial set of conditions and core features	Scope narrowed to 5 conditions, 2 visualization types and a limited AI assistant
Connect AI requirements to safety and source-citation tests	Test set of 30 normal and 20 crisis/diagnosis questions with source checking
Finalize the user-study methodology	Short discussions and a before/after online survey
Link YouTrack tasks with GitHub commits	Task ID is used in branch names and commit messages

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